Price wedge anomaly signals: construction log

September 12, 2026

How to read this

One entry per characteristic. *Agreement* is the mean correlation, across the ten deciles and 462 formation months, between our decile capitalisation weights and the replication package's own (`datas_all_charP`). It tests the characteristic and the sort alone, independently of the 241-month tracking and the discount factor. Above 0.85 the construction is treated as verified. *Error* is our price wedge less the paper's Table 1 value, in percentage points.

Rejected alternatives records what was tried and did not work, so the same ground is not covered twice.

Summary

Signal	Group	Agreement	Long err.	Short err.	Status
OA	intangibles	0.963	-1.53	0.15	verified
AOA	intangibles	0.971	-1.15	0.05	verified
ATO	profitability	0.974	-0.86	0.16	verified
RNA	profitability	0.978	0.60	-0.30	verified
dSOUT	investment	0.992	-0.53	-0.68	verified
dPIA	investment	0.943	0.23	-0.87	verified
aSIZE	trading frictions	0.856	-14.69	0.21	verified
A2ME	value	0.972	3.57	-0.13	verified
AT	trading frictions	0.988	-0.02	0.11	verified
BEME	value	0.977	1.24	-0.70	verified
C2A	value	0.963	-0.06	-0.20	verified
C2D	value	0.959	-0.48	-2.03	verified
CAT	profitability	0.977	1.23	-0.32	verified
D2P	value	0.964	0.47	-0.36	verified
dCEQ	investment	0.951	-2.79	-1.51	verified
dGS	profitability	0.966	-0.69	-0.25	verified
dSO	value	0.966	-0.40	-0.53	verified
E2P	value	0.978	3.00	-0.41	verified
EPS	value	0.977	0.82	-3.88	verified
IDIOV	trading frictions	0.955	-0.21	-0.99	verified
I2A	investment	0.971	0.01	-1.30	verified
IPM	profitability	0.962	-0.78	-1.43	verified

Signal	Group	Agreement	Long err.	Short err.	Status
IVC	investment	0.970	-0.98	0.26	verified
SIZE	trading frictions	0.997	0.09	-0.21	verified
NOA	investment	0.967	-1.12	-0.26	verified
OL	intangibles	0.959	1.35	-0.16	verified
BETA_d	trading frictions	0.952	0.05	-0.17	verified
PCM	profitability	0.977	-0.44	-1.57	verified
PM	profitability	0.973	-0.61	-1.53	verified
PROF	profitability	0.978	-0.53	-0.29	verified
Q	value	0.978	0.91	-0.21	verified
MAXRET	trading frictions	0.984	-0.22	-1.16	verified
ROA	profitability	0.965	-0.06	0.53	verified
ROC	profitability	0.972	-0.95	0.24	verified
ROE	profitability	0.972	-0.41	-2.69	verified
ROIC	profitability	0.964	0.07	-0.23	verified
R_12_2	past returns	0.979	0.17	0.29	verified
R_12_7	past returns	0.984	-0.13	-0.43	verified
R_6_2	past returns	0.978	-0.02	-0.91	verified
R_2_1	past returns	0.986	-0.35	-0.45	verified
R_36_13	past returns	0.988	-0.95	-0.40	verified
S2C	value	0.968	0.15	-0.92	verified
S2P	value	0.979	5.01	-0.62	verified
SG	value	0.963	0.12	-0.82	verified
SAT	profitability	0.971	1.65	-0.20	verified
SPREAD	trading frictions	0.964	-1.48	2.38	verified
sdTURN	trading frictions	0.991	0.03	-0.06	verified
sdDVOL	trading frictions	0.896	-5.28	-0.08	verified
TAN	intangibles	0.978	-1.35	-1.38	verified
RETVOL	trading frictions	0.988	-0.14	-0.32	verified
DP	value	0.841	1.39	-3.39	unresolved
TNOVR	trading frictions	0.790	0.25	1.68	unresolved
SUV	trading frictions	0.073	3.76	-3.27	unresolved
DTO	trading frictions	0.363	1.37	-1.79	unresolved
aBEME	value	0.510	-0.81	-7.58	unresolved
aSAT	profitability	0.447	-3.51	3.04	unresolved
aPM	profitability	0.284	1.62	1.08	unresolved

Construction, signal by signal

OA — Operating accruals

Group intangibles **Source** Sloan (1996), per Appendix Table A.1 **Frequency** annual **Agreement** 0.963

Inputs. `act, che, lct, dlc, txp, dp, at`

Formula.

> OA = (d act - d che - (d lct - d dlc - d txp - dp)) / at[t-1]
> where d txp treats a missing value as zero.

Note the placement of dp: it sits INSIDE the bracket, so expanding gives OA = d act - d che - d lct + d dlc + d txp + dp, i.e. depreciation is ADDED to accruals.

Notes. This is not Sloan (1996) and not what Table A.1's own wording says ("changes in non-cash working capital minus depreciation"). It is nevertheless unambiguously what the paper's data does: netting depreciation out gives agreement 0.286, adding it gives 0.963, and decile one's share of market capitalisation comes out at 5.29% against a published 5.30%. Chen and Zimmermann's PctAcc.do carries the identical bracket placement while their Accruals.do does not, which suggests a shared transcription. WORTH RAISING WITH THE COAUTHORS: as published, OA and AOA are not accruals net of depreciation, so capital-intensive firms sort toward the high-accrual leg because of their depreciation rather than away from it.

Rejected alternatives.

Variant	Agreement	Verdict
Sloan's grouping with dp subtracted outside the bracket	0.286	rejected
literal reading of A.1, all three current-liability terms subtracted	0.065	rejected
dp subtracted, scaled by average rather than lagged total assets	0.251	rejected
dp subtracted, every delta zero-filled	0.163	rejected
change in net operating assets	0.234	rejected
change in working capital from wcap directly	-0.058	rejected
cash-flow statement form, ni less oancf, over average assets	0.087	rejected
cash-flow form with balance-sheet fallback	0.269	rejected
scaled by sales, by absolute earnings, by current assets	0.154	rejected
winsorised 1/99 monthly, or —OA— capped at 1, 0.5, 0.3	0.065	no effect; outliers are not the cause
JKP oaccruals_at taken directly from their library	0.278	better than ours was, worse than the fix
shifted +/- 6 months against the published series	–	no peak; not a timing offset

AOA — Absolute operating accruals

Group intangibles **Source** Bandyopadhyay, Huang and Wirjanto (2010) **Frequency** annual **Agreement** 0.971

Inputs. `act, che, lct, dlc, txp, dp, at`

Formula.

AOA = —OA—, with OA exactly as above (depreciation added, not subtracted).

Notes. Follows OA mechanically. Neither Chen and Zimmermann nor Jensen, Kelly and Pedersen carry a Bandyopadhyay signal, so there is no third implementation to compare against; it is verified only through OA.

Rejected alternatives.

Variant	Agreement	Verdict
absolute value of every rejected OA variant above	0.087	rejected

ATO — Asset turnover

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.974

Inputs. `sale, at, che, ivao, dlc, dltt, mib, pstk, ceq`

Formula.

operating assets = at - che - ivao (che, ivao missing -¿ 0)
operating liabilities = at - dlc - dltt - mib - pstk - ceq
(dlc, dltt, mib, pstk missing -¿ 0; ceq REQUIRED)
net operating assets = operating assets - operating liabilities
ATO = sale / NOA[t-1]

Notes. Common equity must be required rather than zero-filled. Filled, operating liabilities collapse to assets less debt, net operating assets go to nearly zero, and anything divided by them explodes into the end deciles. Agreement 0.78 filled against 0.92 required. This is Chen and Zimmermann's convention in RetNOA.do, where dlc, dltt, mib and pstk are zero-filled and ceq is not.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled like the other liability terms	0.780	rejected
contemporaneous rather than lagged net operating assets	0.582	rejected
operating assets without subtracting ivao	0.558	rejected
observations with non-positive NOA discarded	0.494	rejected
Soliman's component-built operating assets (rect+invt+aco+ppent+intan-ap-lco-lo), averaged	0.241	rejected
lagged sales over lagged NOA	0.625	rejected

RNA — Return on net operating assets

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.978

Inputs. `oiadp, at, che, ivao, dlc, dltt, mib, pstk, ceq`

Formula.

RNA = oiadp / NOA[t-1], with NOA exactly as defined under ATO
(ceq required, the other liability terms zero-filled).

Notes. Same common-equity point as ATO: 0.76 with ceq filled, 0.95 required.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled	0.758	rejected
contemporaneous NOA	0.690	rejected
operating assets without ivao	0.551	rejected
non-positive NOA discarded	0.538	rejected
Soliman component-built operating assets	0.411	rejected

dSOUT — Change in shares outstanding

Group investment **Source** Pontiff and Woodgate (2008) **Frequency** monthly **Agreement** 0.992
Inputs. `shrout, mthcumfacshr`

Formula.

adjusted shares s[t] = shrout[t] * mthcumfacshr[t]
dSOUT[t] = 100 * (s[t] / s[t-12] - 1)

Notes. CRSP's share counts are not split-adjusted, so raw growth reads every split as a 100% issuance. The cumulative share factor is the exact adjustment.

Rejected alternatives.

Variant	Agreement	Verdict
raw shrout growth, unadjusted for splits	0.513	rejected
split adjustment backed out of market cap and the ex-dividend return	0.883	close, superseded

dPIA — Change in plant and inventory over assets

Group investment **Source** Lyandres, Sun and Zhang (2008) **Frequency** annual **Agreement** 0.943
Inputs. `ppegt, invt, at`

Formula.

dPIA = (d ppegt + d invt) / at[t-1],
where each delta treats a missing component as no change (zero) rather than discarding the firm.

Rejected alternatives.

Variant	Agreement	Verdict
both deltas required non-missing	0.777	rejected
scaled by average rather than lagged assets	0.771	rejected
net rather than gross plant (ppent)	0.493	rejected
scaled by lagged (ppegt + invt)	0.420	rejected

aSIZE — Industry-adjusted size

Group trading frictions **Source** Asness, Porter and Stevens (2000) **Frequency** monthly **Agreement** 0.856

Inputs. `cap, siccd`

Formula.

aSIZE = SIZE - mean(SIZE) within the Fama-French 48 industry that month, equal-weighted over ordinary common shares.

Notes. Just above the threshold at 0.85, but its wedge is still 13 points from the paper's on the long leg – the widest gap of any characteristic that passes the construction check. Treat as unresolved in practice.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 10 and 30 industries	0.866	marginally better than 48, within noise
industry median instead of mean	0.877	marginally better, not adopted
log size demeaned within industry	0.754	rejected
partial adjustment at 0.75	0.872	marginally better, not adopted
capitalisation-weighted industry mean	0.317	rejected

A2ME — Total assets (at) over market capitalization (prc x shrout)

Group value **Source** bhandari1988 **Frequency** monthly **Agreement** 0.972

Inputs. `at`

Formula.

at / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.700	rejected

AT — Total assets (at)

Group trading frictions **Source** gandhi2015 **Frequency** annual **Agreement** 0.988

Inputs. `at`

Formula.

Total assets.

BEME — Book equity to market equity. Book equity is shareholders' equity (seq

Group value **Source** – **Frequency** monthly **Agreement** 0.977
Inputs. `at, ceq, lt, prc, pstk, pstkl, pstrkrv, seq, shrout, txditc`
Formula.

be / me, with negative book equity discarded before sorting.
Book equity = seq, else ceq+pstk, else at-lt; plus txditc; minus the first available of pstkrv, pstkl, pstk. Market equity is CRSP mthcap for the CURRENT month, in millions.

Notes. FNW state market equity as of December t-1 explicitly. The package's own stored panel says otherwise once its one-month row offset is accounted for: lined up correctly the current month matches 94.9% of firm-months exactly and December 3.9%. Implementing December drops agreement from 0.97 to about 0.70.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December of the prior year	0.700	rejected
market equity lagged one month	0.840	rejected; an artifact of a row-offset error in the comparison
market equity summed across share classes of the same permco	–	marginally better on values (95.7% vs 94.9%), not adopted

C2A — Cash and short-term investments (che) to total assets (at)

Group value **Source** palazzo2012 **Frequency** annual **Agreement** 0.963
Inputs. `at, che`
Formula.

che / at.

C2D — Cashflow to debt. Cashflow is the sum of income and extraordinary item

Group value **Source** ou1989 **Frequency** annual **Agreement** 0.959
Inputs. `dp, ib, lt`
Formula.

(ib + dp) / lt.

CAT — Sales (sale) to lagged total assets (at)

Group profitability **Source** haugen1996 **Frequency** annual **Agreement** 0.977
Inputs. `at, sale`
Formula.

sale / at[t-1].

D2P — Debt to price. Debt is long-term debt (dltt) plus debt in current liab

Group value **Source** litzenberger1979 **Frequency** monthly **Agreement** 0.964
Inputs. `dlc, dltt, prc, shrout`
Formula.

(dltt + dlc) / me, each debt term missing treated as zero.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.710	rejected

dCEQ — Annual % change in book value of equity (ceq)

Group investment **Source** richardson2005 **Frequency** annual **Agreement** 0.951
Inputs. `ceq`
Formula.

Percent change in ceq.

dGS — % change in gross margin minus % change in sales (sale). Gross margin

Group profitability **Source** abarbanell1997 **Frequency** annual **Agreement** 0.966
Inputs. `cogs, sale`
Formula.

Percent growth in gross margin (sale-cogs) less percent growth in sale.

dSO — Log change in the product of shares outstanding (csho) and the adjustm

Group value **Source** fama2008 **Frequency** annual **Agreement** 0.966
Inputs. `ajex, csho`

Formula.

> log(csho*ajex / (csho*ajex)[t-1]), i.e. growth in split-adjusted Compustat share count.

E2P — Income before extraordinary items (ib) to market capitalization (prc x

Group value **Source** basu1983 **Frequency** monthly **Agreement** 0.978
Inputs. `ib`

Formula.

> ib / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.690	rejected

EPS — Income before extraordinary items (ib) to shares outstanding (shrout)

Group value **Source** basu1977 **Frequency** monthly **Agreement** 0.977
Inputs. `ib, shrout`

Formula.

> ib / (shrout/1000), i.e. earnings per share on CRSP's share count.

IDIOV — Standard deviation of the residuals from a regression of excess return

Group trading frictions **Source** ang2006 **Frequency** monthly **Agreement** 0.955
Formula.

> Standard deviation of the residuals from a regression of daily excess returns on the three Fama-French factors, within the month, at least 15 days.

Notes. Emphatically monthly, unlike BETA_d: 0.93 at one month against 0.58 at twelve. JKP corroborate, carrying ivol_ff3_21d at a 21-day window, and our series correlates 0.997 with theirs.

Rejected alternatives.

Variant	Agreement	Verdict
estimated over twelve months	0.582	rejected

I2A — Annual % change in total assets (at)

Group investment **Source** – **Frequency** annual **Agreement** 0.971
Inputs. `at`
Formula.

Percent change in at.

IPM — Pre-tax income (pi) over sales (sale)

Group profitability **Source** – **Frequency** annual **Agreement** 0.962
Inputs. `pi, sale`
Formula.

pi / sale.

IVC — Annual change in inventories (invt) in the last two fiscal years over

Group investment **Source** thomas2002 **Frequency** annual **Agreement** 0.970
Inputs. `at, invt`
Formula.

(invt - invt[t-1]) / ((at + at[t-1])/2).

SIZE — Price (prc) times shares outstanding (shrout)

Group trading frictions **Source** fama1992 **Frequency** monthly **Agreement** 0.997
Inputs. `prc, shrout`
Formula.

Market equity, CRSP mthcap for the current month, in millions.

NOA — Operating assets minus operating liabilities to lagged total assets (a

Group investment **Source** hirshleifer2004 **Frequency** annual **Agreement** 0.967
Inputs. `at, ceq, che, dlc, dltt, ivao, mib, pstk`
Formula.

(operating assets - operating liabilities) / at[t-1], with operating assets and liabilities as defined under ATO.

Notes. Unaffected by the common-equity question that matters so much for ATO and RNA, because NOA uses the level rather than dividing by it.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled	0.917	no effect

OL — Sum of cost of goods sold (cogs) and selling, general and administrati

Group intangibles **Source** novy2011 **Frequency** annual **Agreement** 0.959
Inputs. `at, cogs, xsga`
Formula.

(cogs + xsga) / at, each missing term treated as zero.

BETA_d — The sum of the regression coefficients of daily excess returns on mark

Group trading frictions **Source** lewellen2006 **Frequency** monthly **Agreement** 0.952
Formula.

Sum of the coefficients on the market and its lag, from a regression of daily excess returns on both, estimated over TWELVE months of daily data.

Notes. Twelve months, not one. A one-month Dimson beta is too noisy to sort on: 0.35 at one month, 0.62 at six, 0.94 at twelve, 0.63 at twenty-four. The moments are additive, so a twelve-month cross-product matrix is the sum of twelve monthly ones.

Rejected alternatives.

Variant	Agreement	Verdict
estimated over one month	0.353	rejected
six months	0.617	rejected
twenty-four and sixty months	0.631	rejected

PCM — Net sales (sale) minus cost of goods sold (cogs) all scaled by net sal

Group profitability **Source** gorodnichenko2016 **Frequency** annual **Agreement** 0.977
Inputs. `cogs, sale`
Formula.

(sale - cogs) / sale.

PM — Operating Income after depreciation (oiadp) to sales (sale)

Group profitability **Source** soliman2008 **Frequency** annual **Agreement** 0.973
Inputs. `oiadp, sale`
Formula.

oiadp / sale.

Rejected alternatives.

Variant	Agreement	Verdict
net income over revenue, as Chen and Zimmermann implement Soliman	0.387	rejected
oiadp over average sales	0.732	rejected

PROF — Gross profitability (gp) over book equity as defined in BEME

Group profitability **Source** ball2015 **Frequency** annual **Agreement** 0.978
Inputs. `gp`
Formula.

gp / be, with book equity as defined under BEME.

Q — Total assets (at) plus market value of equity (shrout x prc) minus boo

Group value **Source** – **Frequency** monthly **Agreement** 0.978
Inputs. `at, ceq, txdb`
Formula.

(at + me - ceq - txdb) / at, txdb missing treated as zero, me the current month's market equity in millions.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December of the prior year	0.610	rejected

MAXRET — Maximum daily return in the previous month

Group trading frictions **Source** bali2011 **Frequency** monthly **Agreement** 0.984
Formula.

> Maximum daily return within the month.

Notes. Correlates 1.000 with JKP's rmax1_21d.

ROA — Income before extraordinary items (ib) to lagged total assets (at)

Group profitability **Source** balakrishnan2010 **Frequency** annual **Agreement** 0.965
Inputs. `at, ib`
Formula.

> ib / at[t-1].

ROC — Market value of equity (shrout x prc) plus long-term debt (dltt) minus

Group profitability **Source** chandrashekar2009 **Frequency** monthly **Agreement** 0.972
Inputs. `at, che, dltt`
Formula.

> (me + dltt - at) / che.

ROE — Income before extraordinary items (ib) to lagged book-value of equity

Group profitability **Source** haugen1996 **Frequency** annual **Agreement** 0.972
Inputs. `ib`
Formula.

> ib / be[t-1].

ROIC — Earnings before interest and taxes (ebit) less non-operating income (n

Group profitability **Source** brown2007 **Frequency** annual **Agreement** 0.964
Inputs. `ceq, che, ebit, lt, nopi`
Formula.

> (ebit - nopi) / (ceq + lt - che), nopi missing treated as zero.

Notes. Appendix Table A.1 words the denominator as a sum, which would add cash to invested capital. Subtracting it is both what invested capital means and what the data says: 0.64 adding against 0.93 subtracting.

Rejected alternatives.

Variant	Agreement	Verdict
cash added to invested capital, as A.1 words it	0.640	rejected
every component required non-missing	0.640	no effect

R_12_2 — Cumulative return from 12 months to 2 months ago

Group past returns **Source** fama1996 **Frequency** monthly **Agreement** 0.979

Formula.

> Compound return over months t-11..t-1, stored on row t.

Notes. Stored ONE MONTH EARLIER than the name suggests. The sort's own one-month lag then produces the stated twelve-to-two window. Storing the literal window drops agreement from 0.97 to 0.65.

Rejected alternatives.

Variant	Agreement	Verdict
literal window, months t-12..t-2	0.647	rejected

R_12_7 — Cumulative return from 12 months to 7 months ago

Group past returns **Source** novy2012 **Frequency** monthly **Agreement** 0.984

Formula.

> Compound return over months t-11..t-6, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-12..t-7	0.533	rejected

R_6_2 — Cumulative return from 6 months to 2 months ago

Group past returns **Source** jegadeesh1993 **Frequency** monthly **Agreement** 0.978

Formula.

> Compound return over months t-5..t-1, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-6..t-2	0.507	rejected

R_2_1 — Lagged one month return

Group past returns **Source** jegadeesh1990 **Frequency** monthly **Agreement** 0.986
Formula.

The CURRENT month's return, stored on row t.

Notes. The one-month shift matters most here: with a single month there is no overlap to rescue a timing error, and the literal 'lagged one-month return' gives 0.11 against 0.97.

Rejected alternatives.

Variant	Agreement	Verdict
lagged one-month return, as A.1 words it	0.108	rejected

R_36_13 — Cumulative return from 36 months to 13 months ago

Group past returns **Source** de1985 **Frequency** monthly **Agreement** 0.988
Formula.

Compound return over months t-35..t-12, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-36..t-13	0.744	rejected

S2C — Net sales (sale) to cash and short-term investments (che)

Group value **Source** ou1989 **Frequency** annual **Agreement** 0.968
Inputs. `che, sale`
Formula.

sale / che.

S2P — Net sales (sale) to market capitalization (shrout x prc)

Group value **Source** lewellen2015 **Frequency** monthly **Agreement** 0.979

Inputs. `sale`

Formula.

sale / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December	0.720	rejected

SG — % growth rate in sales (sale)

Group value **Source** lakonishok1994 **Frequency** annual **Agreement** 0.963
Inputs. `sale`

Formula.

Percent change in sale.

SAT — Sales (sale) to total assets (at)

Group profitability **Source** soliman2008 **Frequency** annual **Agreement** 0.971
Inputs. `at, sale`

Formula.

sale / at.

Rejected alternatives.

Variant	Agreement	Verdict
sale over average assets	0.786	rejected
sale over lagged assets	0.736	rejected

SPREAD — Average daily bid-ask spread in the previous month

Group trading frictions **Source** chung2014 **Frequency** monthly **Agreement** 0.964
Formula.

Mean over the month of the daily high-low range, 2*(high-low)/(high+low), requiring at least 15 days.

Notes. NOT the quoted bid-ask spread, although FNW cite Chung and Zhang (2014). CRSP carries quotes for 9% of daily rows in the 1960s and 53% in the 1980s, so a quoted spread cannot be computed over this sample: 0.02 against 0.94.

Rejected alternatives.

Variant	Agreement	Verdict
quoted bid-ask spread, 2*(ask-bid)/(ask+bid)	0.017	rejected
high-low range over the closing price	0.933	marginally worse

sdTURN — Standard deviation of residuals from a regression of daily turnover on

Group trading frictions **Source** chordia2001 **Frequency** monthly **Agreement** 0.991
Inputs. `shrout, vol`

Formula.

> Standard deviation within the month of daily turnover (volume over shares outstanding), requiring at least 15 days.

Rejected alternatives.

Variant	Agreement	Verdict
coefficient of variation of turnover (Chordia et al.)	0.004	rejected

sdDVOL — Standard deviation of residuals from a regression of daily volume (vol

Group trading frictions **Source** chordia2001 **Frequency** monthly **Agreement** 0.896
Inputs. `vol`

Formula.

> Standard deviation within the month of log(1 + daily dollar volume), requiring at least 15 days.

Notes. In LOGS. In levels the dispersion measures scale rather than variability: 0.06 against 0.87.

Rejected alternatives.

Variant	Agreement	Verdict
standard deviation of dollar volume in levels	0.057	rejected
standard deviation of log share volume	0.632	rejected
coefficient of variation of dollar volume (Chordia et al.)	0.521	rejected

TAN — Tangibility is defined as (0.715 x total receivables (rect) + 0.547 x

Group intangibles **Source** hahn2009 **Frequency** annual **Agreement** 0.978
Inputs. `at, che, invt, ppent, rect`

Formula.

(0.715*rect + 0.547*invt + 0.535*ppent + che) / at, every component REQUIRED.

Notes. Zero-filling a missing component reads as 'this firm holds no tangible assets' and sends firms with incomplete filings to the bottom decile: 0.48 against 0.96.

Rejected alternatives.

Variant	Agreement	Verdict
missing components filled with zero	0.478	rejected

RETVOL — Standard deviation of residuals from a regression of excess returns on

Group trading frictions **Source** ang2006 **Frequency** monthly **Agreement** 0.988
Formula.

Standard deviation of daily excess returns within the month, at least 15 days.

Notes. Correlates 1.000 with JKP's rvol_21d.

DP — Dividend yield

Group value **Source** Litzenberger and Ramaswamy (1979) **Frequency** monthly **Agreement** 0.841

Inputs. `ret, retx, cap`

Formula.

dollars paid in month s = (ret[s] - retx[s]) * cap[s-1]
DP[t] = sum of those dollars over months t-11..t, divided by cap[t],
requiring at least six of the twelve months.

Notes. Dividends in dollars over market capitalisation, not dividends per share over price. Per share the share count drifts through the year and agreement falls to 0.841. JKP's div12m_me, which is this same quantity, scores 0.928 against the package, and building it natively gives 0.924.

Rejected alternatives.

Variant	Agreement	Verdict
dividends per share over price (twelve-month sum)	0.841	rejected
per-share dividends restated for splits before summing	0.735	rejected; the more coherent calculation agrees worse
sum of twelve monthly dividend yields, no price	0.743	rejected

Variant	Agreement	Verdict
cash distributions from crsp.stkdistributions over price	0.348	rejected despite being the more principled source
dollar dividends over lagged rather than current market cap	0.869	rejected
JKP div_me (one month), div6m_me, div_at	0.652	rejected

TNOVR — Share turnover

Group trading frictions **Source** Datar, Naik and Radcliffe (1998) **Frequency** monthly **Agreement** 0.790

Inputs. `vol, shrout`

Formula.

TNOVR[t] = (vol[t] + vol[t-1] + vol[t-2]) / (3 * shrout[t])

Notes. Averaged over three months. Table A.1 reads as one month and the source paper does not say; three tracks the published weights best. Still below the verification threshold at 0.79, so this one is not settled.

Rejected alternatives.

Variant	Agreement	Verdict
single month, vol/shrout	0.707	rejected
twelve-month average	0.699	rejected
mean daily turnover within the month	0.706	rejected
Nasdaq volume scaled down 50% pre-1997 and 38% after	0.679	rejected
dollar volume over market cap	0.708	rejected
JKP turnover_126d	0.713	rejected

SUV — Standardised unexplained volume

Group trading frictions **Source** Garfinkel (2009) **Frequency** monthly (daily inputs) **Agreement** 0.073

Inputs. `dlyvol, dlyret`

Formula.

Within a month, regress log(1 + daily volume) on a constant and the
absolute values of positive and negative daily returns:
ln(1+v_i) = a + b*max(r_i,0) + c*—min(r_i,0)— + e_i
Estimated on the previous month and applied to the current one:
SUV[t] = (sum_i ln(1+v_i) - predicted) / (n * sd(e))
with predicted and sd(e) from month t-1's fit. At least 15 days required.

Notes. UNRESOLVED at 0.07. Neither Chen and Zimmermann nor JKP carry a Garfinkel signal, so there is no reference implementation anywhere. The level of the decile weights comes out close (8.85 against a published 8.94 on decile one) while the month-to-month variation does not, which is the signature of a measure that is right in construction but wrong in timing or in which residual is taken. Full daily data has been pulled to test day-level variants the monthly moments cannot express.

Rejected alternatives.

Variant	Agreement	Verdict
regression in volume levels rather than logs	0.080	rejected
estimation window of 1, 6, 12 and 24 months applied out of sample	0.092	rejected at every length
previous month taken on a dense grid so the lag cannot cross a gap	0.072	rejected
in-sample residual dispersion rather than an out-of-sample residual	0.002	rejected
standardised by sqrt(n) rather than n	0.073	rejected

DTO — Detrended turnover

Group trading frictions **Source** Garfinkel (2009) **Frequency** monthly (daily inputs) **Agreement** 0.363

Inputs. `dlyvol, shrout, primaryexch`

Formula.

> daily turnover, with Nasdaq volume scaled by 0.50 before 1997 and 0.38 after, less the cross-sectional mean that day, then detrended by subtracting its own median over the previous 180 trading days.

Notes. UNRESOLVED at 0.36, and the cause is known: the 180-trading-day median is not a moment of any month, so it cannot come out of the server-side aggregation the other eight daily characteristics use. It is currently approximated by a rolling median of nine monthly means. Full daily data has been pulled to compute the true median.

Rejected alternatives.

Variant	Agreement	Verdict
rolling median over 6, 9 and 12 months of monthly means	0.367	best available approximation so far

aBEME — Industry-adjusted book to market

Group value **Source** Asness, Porter and Stevens (2000) **Frequency** monthly **Agreement** 0.510

Inputs. `be, cap, siccd`

Formula.

> aBEME = BEME - mean(BEME) within the firm's Fama-French 48 industry that month, equal-weighted over ordinary common shares that have BEME.

Notes. UNRESOLVED at 0.51, and the puzzle is that its parent BEME verifies at 0.97, so the adjustment itself is what fails. Our decile one matches (3.93 against a published 3.77) while decile ten does not (13.13 against 18.30): the published adjustment removes far less of the growth tilt than ours does. The search below has been exhaustive and nothing closes it. Most likely the industry classification comes from a source we do not have – plausibly Baba-Yara, Boons and Tamoni's own file, since the paper says the characteristic set comes from that project as well as from FNW.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 5, 10, 12, 17, 30, 38, 48 and 49 industries	0.521	ff30 marginally best, none close
one-digit, two-digit and three-digit SIC	0.477	rejected
Compustat historical SIC (sich), with and without CRSP fall-back	0.506	rejected
Compustat header SIC from comp.company	0.468	rejected
a firm's modal SIC over its life	0.509	rejected
industry median instead of mean	0.511	rejected
capitalisation-weighted industry mean	0.227	rejected
industry mean over all of CRSP rather than common shares only	0.099	rejected
industry mean over NYSE firms only	0.384	rejected
industry mean over the larger half of each industry	0.401	rejected
ratio to the industry mean, and log of that ratio	0.440	rejected
standardised within industry (divided by the industry sd)	0.348	rejected
percentile rank within industry	0.421	rejected
cross-sectional rank less the industry mean rank	0.391	rejected
leave-one-out industry mean	0.512	rejected
aggregate industry ratio (sum of book over sum of market)	0.226	rejected
partial adjustment, subtracting 0.25, 0.5 or 0.75 of the industry mean	0.458	rejected
previous year's industry mean	0.343	rejected

aSAT — Industry-adjusted asset turnover

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.447

Inputs. `sale, at, siccd`

Formula.

> aSAT = SAT - mean(SAT) within the Fama-French 48 industry that month, equal-weighted over ordinary common shares.

Notes. UNRESOLVED at 0.45. Parent SAT verifies at 0.90. Same search as aBEME.

Rejected alternatives.

Variant	Agreement	Verdict
the full classification and operation search listed under aBEME	0.448	ff48 demeaning best, none close
aggregate industry ratio (sum of sales over sum of assets)	0.264	rejected
partial adjustment at 0.25, 0.5, 0.75	0.261	rejected

aPM — Industry-adjusted profit margin

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.284
Inputs. `oiadp, sale, siccd`
Formula.

aPM = PM - mean(PM) within the Fama-French 48 industry that month,
equal-weighted over ordinary common shares.

Notes. UNRESOLVED at 0.28, the worst of the industry-adjusted three even though parent PM verifies at 0.97. Separately, its one-month alpha is 0.00012 a month, statistically indistinguishable from zero, so which leg is "long" is not well defined for this characteristic in either dataset. Coverage is also poor in recent years (76% missing in 2024-25) and should be looked at.

Rejected alternatives.

Variant	Agreement	Verdict
the full classification and operation search listed under aBEME	0.340	two-digit SIC marginally best, none close
partial adjustment at 0.25, 0.5, 0.75	0.261	rejected
industry mean over the larger half, and the prior year's mean	0.146	rejected